

Impulse & Momentum – AP-1



Which do you think has more momentum?



Momentum



- Momentum is a measure of how hard it is to stop or turn a moving object.
- What characteristics of an object would make it hard to stop or turn?

Calculating Momentum

- For one particle

$$\mathbf{p} = m\mathbf{v}$$

- **Note that momentum is a vector with the same direction as the velocity!**

- For a system of multiple particles

$$\mathbf{p}_T = \sum \mathbf{p}_i \quad \text{add up the momentum vectors of each object}$$

- The unit of momentum is...
kg m/s or Ns

Sample Problem



- Calculate the momentum of a 65-kg sprinter running east at 10 m/s.

Sample Problem

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$$p = mv = (65 \text{ kg})(10 \text{ m/s}) = 650 \text{ kg m/s}$$



Sample Problem



- Calculate the momentum of a system composed of a 65-kg sprinter running east at 10 m/s and a 75-kg sprinter running west at 10 m/s.



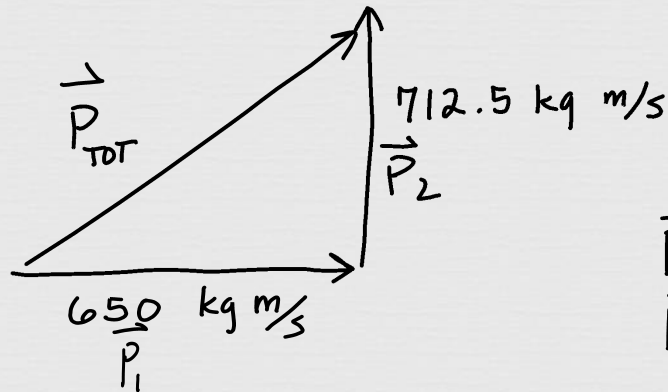
- Calculate the momentum of a system composed of a 65-kg sprinter running east at 10 m/s and a 75-kg sprinter running north at 9.5 m/s.



Sample Problem

- Calculate the momentum of a system composed of a 65-kg sprinter running east at 10 m/s and a 75-kg sprinter running north at 9.5 m/s.

sprinter ① $p = (65 \text{ kg})(10 \text{ m/s}) = 650 \text{ kg m/s}$
sprinter ② $p = (75 \text{ kg})(9.5 \text{ m/s}) = 712.5 \text{ kg m/s}$



$$\theta = \tan^{-1} \left(\frac{712.5}{650} \right)$$
$$\theta = 47.6^\circ$$

$$\vec{P} = \sqrt{650^2 + 712.5^2}$$
$$\vec{P} = 964 \text{ kg m/s} @ 47.6^\circ$$

Change in Momentum



- Like any change, change in momentum is calculated by looking at final and initial momentums.
- $\Delta \mathbf{p} = \mathbf{p}_f - \mathbf{p}_i$
 - $\Delta \mathbf{p}$: change in momentum
 - $\mathbf{p}_f$: final momentum
 - $\mathbf{p}_i$: initial momentum

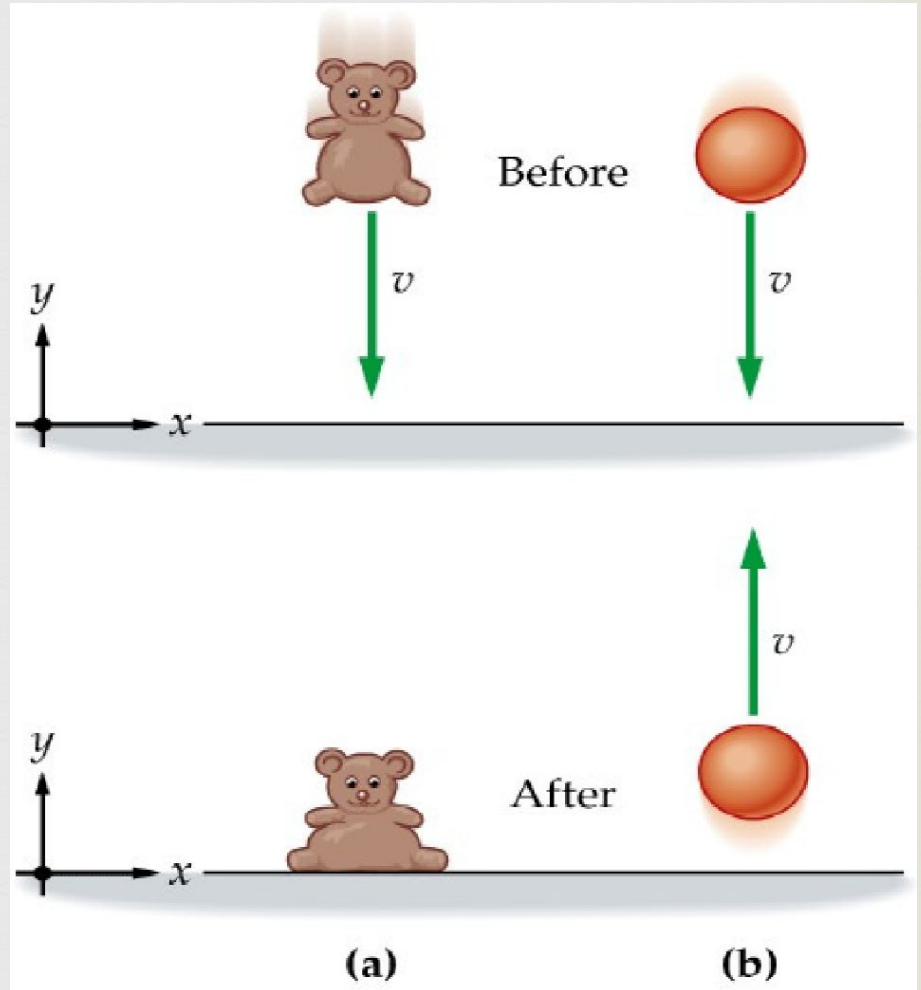
Momentum change demonstration

- Using only a meter stick, find the momentum change of each ball when it strikes the desk from a height of exactly one meter.
- Which ball, Bouncy or Lazy, has the greatest change in momentum?



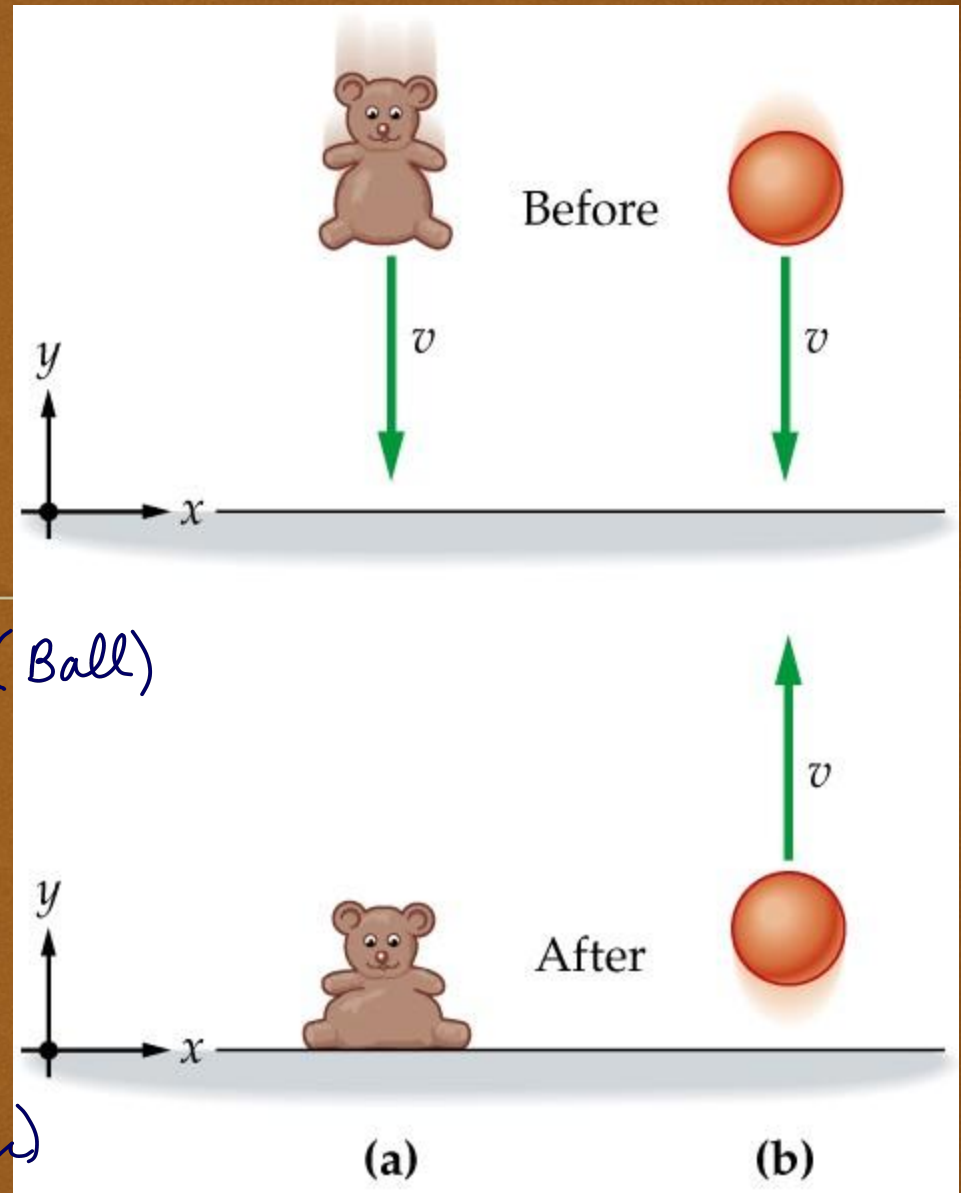
Wording dilemma

- In which case is the magnitude of the momentum change greatest?
- In which case is the change in the magnitude of the momentum greatest?



Wording dilemma

- In which case is the magnitude of the momentum change greatest? (Ball)
- In which case is the change in the magnitude of the momentum greatest? (Teddy Bear)



IMPULSE



Impulse (J)



- Impulse is the product of an external force and time, which results in a change in momentum of a particle or system.
- $J = F t$ and $J = \Delta p$
 - Therefore $Ft = \Delta p$
 - Units: N s or kg m/s (same as momentum)

IMPULSE



| A 57 g tennis ball is served at 45 m/s. If the ball started from rest, what impulse was applied to the ball by the racket?

IMPULSE

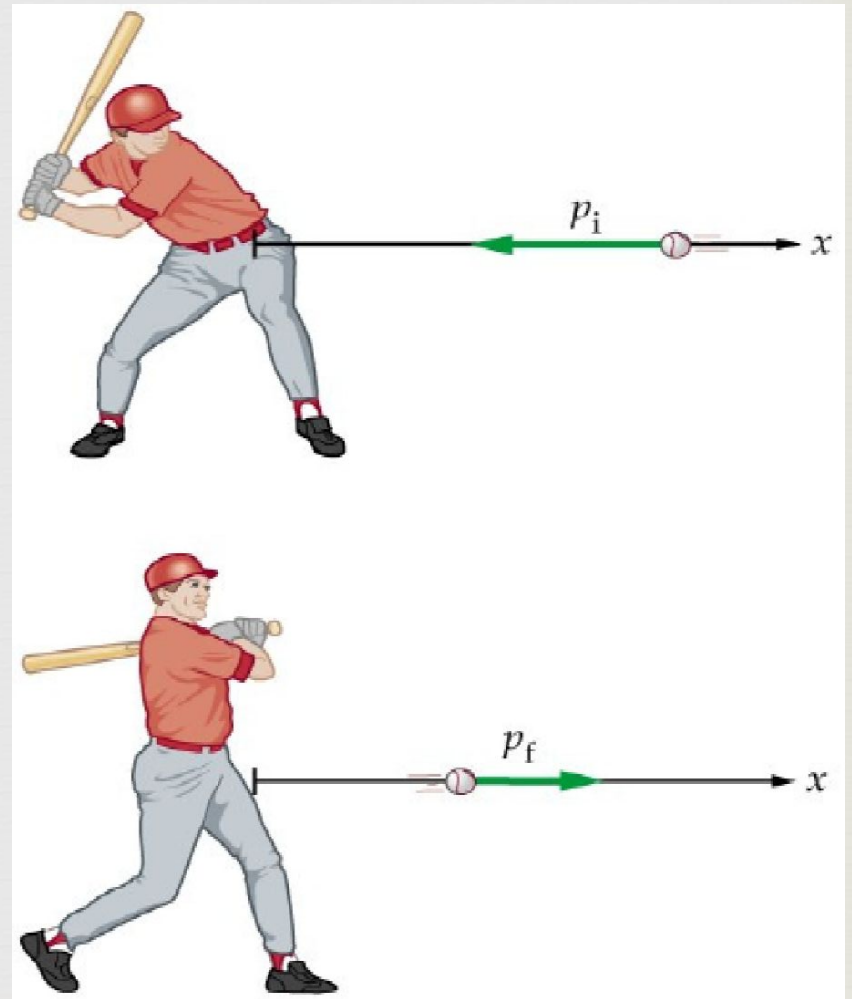


|| In a Little League baseball game, the 145 g ball reaches the batter with a speed of 15.0 m/s. The batter hits the ball, and it leaves his bat with a speed of 20.0 m/s in exactly the opposite direction.

- What is the magnitude of the impulse delivered by the bat to the ball?
- If the bat is in contact with the ball for 1.5 ms, what is the magnitude of the average force exerted by the bat on the ball?

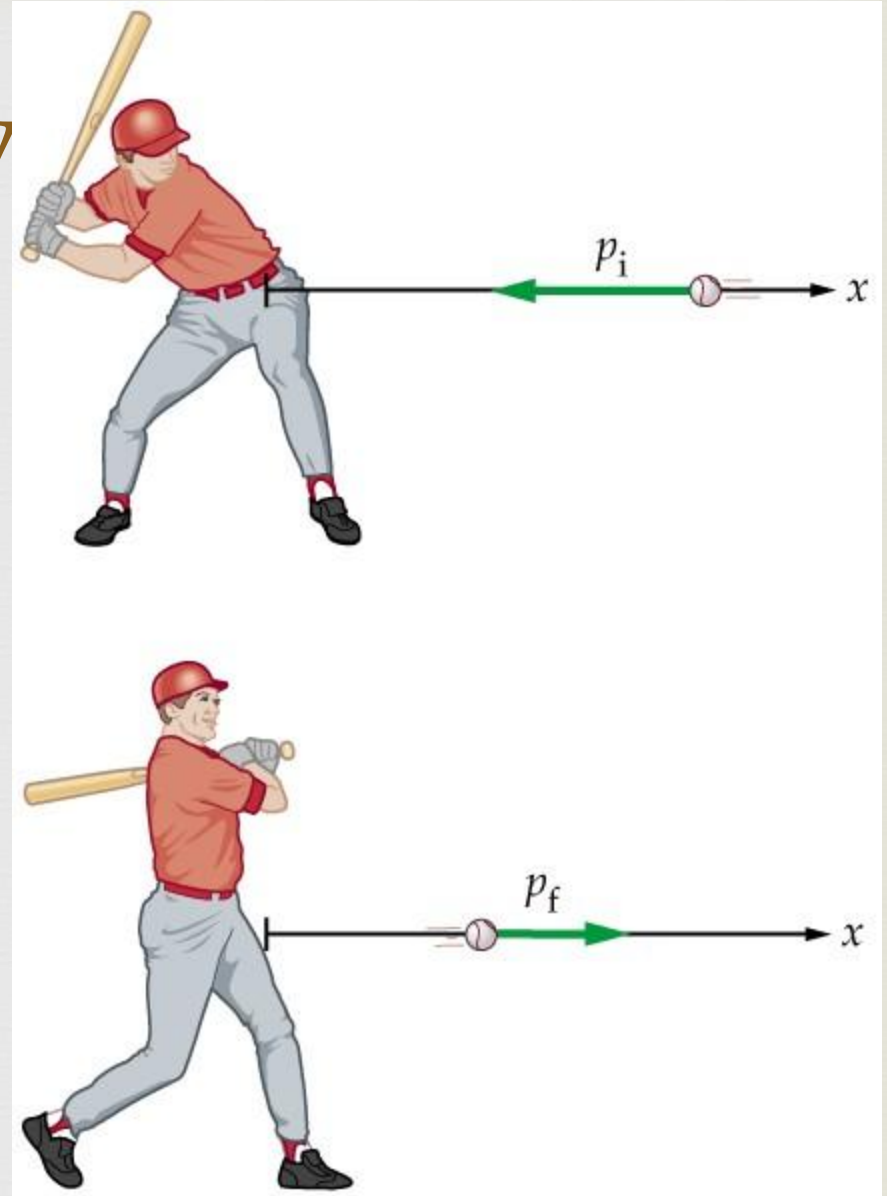
Impulsive Forces

- Usually high magnitude, short duration.
- Suppose the ball hits the bat at 90 mph and leaves the bat at 90 mph, what is the magnitude of the momentum change?
- What is the change in the magnitude of the momentum?

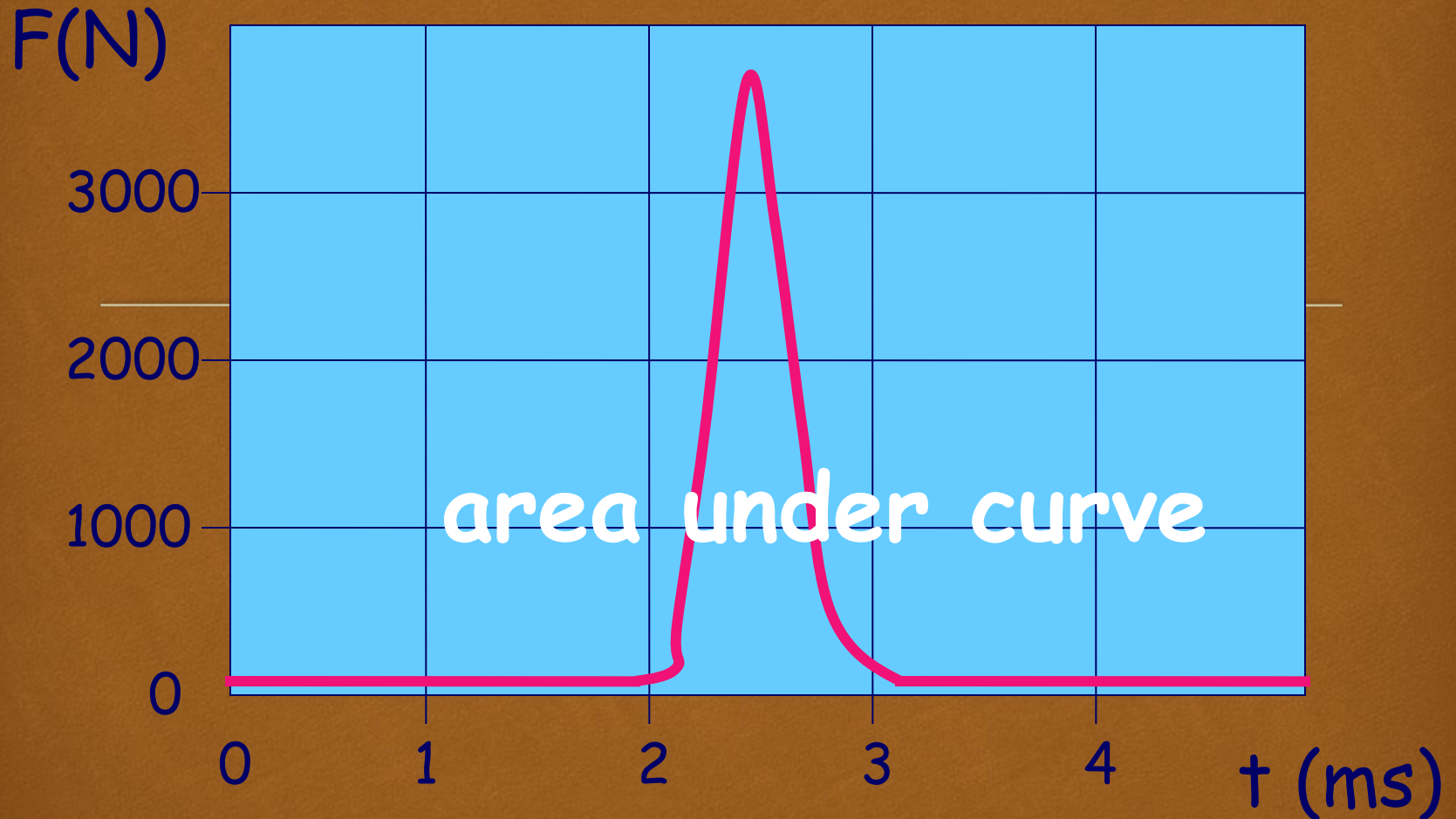


Impulsive

- Usually **high magnitude, short duration**.
- Suppose the ball hits the bat at 90 mph and leaves the bat at 90 mph, what is the magnitude of the momentum change? $2mv$
- What is the change in the magnitude of the momentum? 0

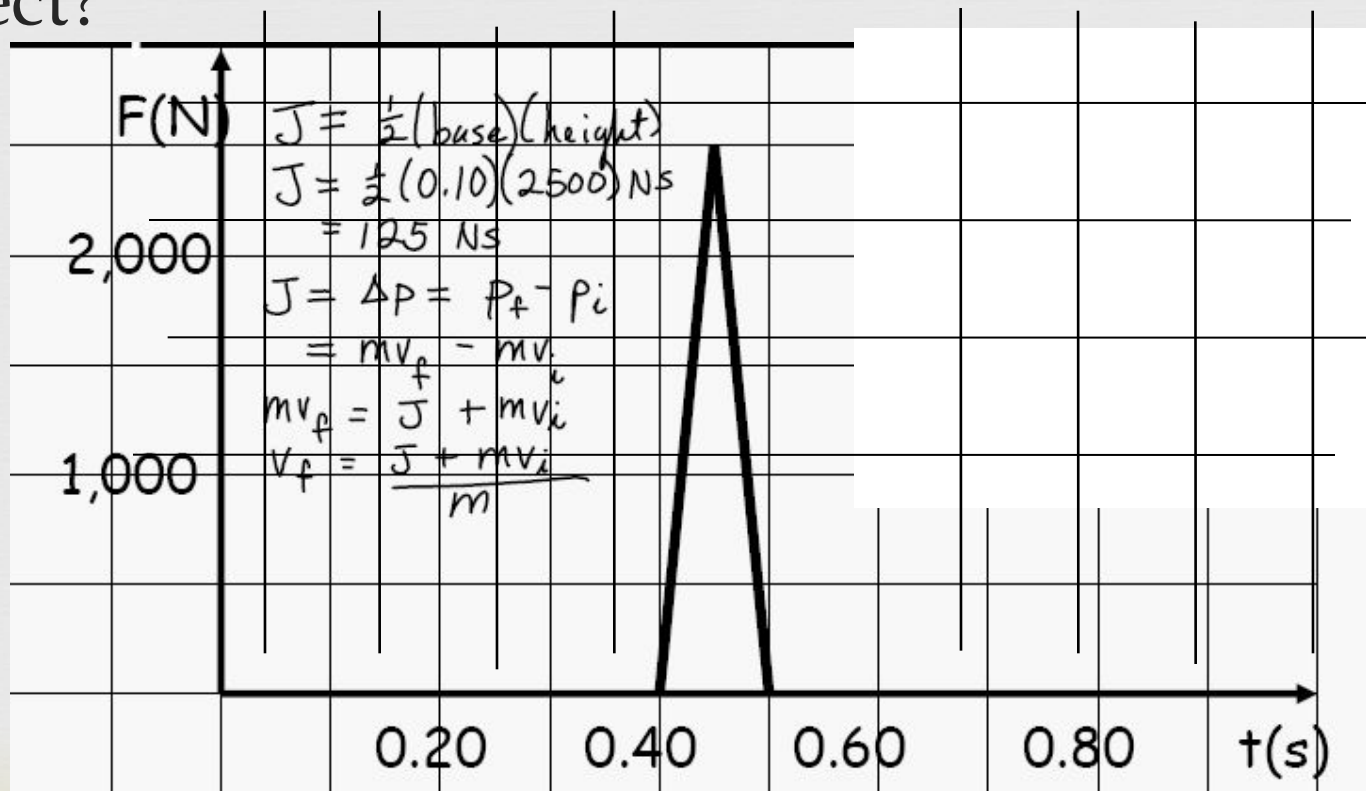


Impulse (J) on a graph

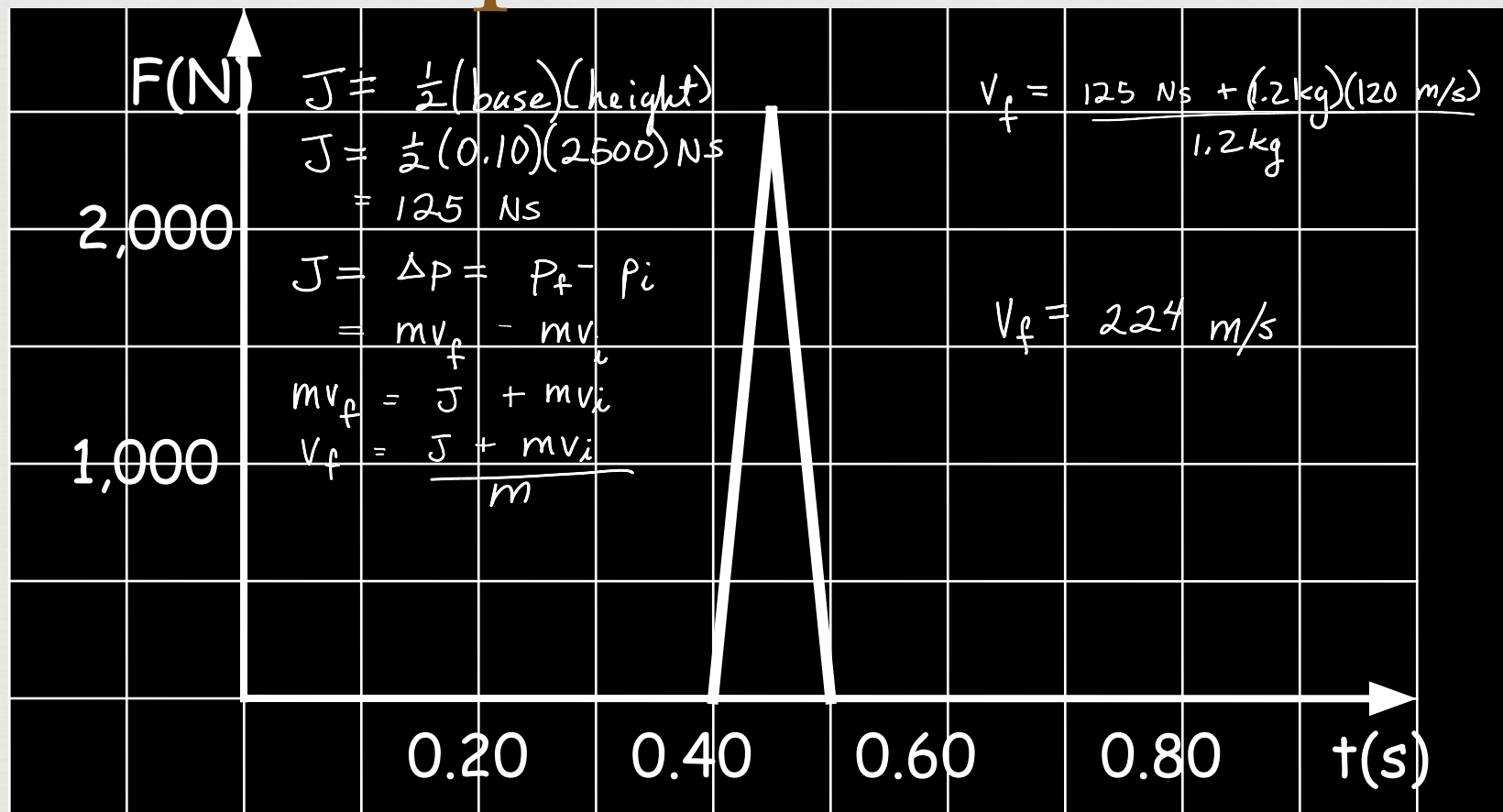


Sample Problem

This force acts on a 1.2 kg object moving at 120.0 m/s. The direction of the force is aligned with the velocity. What is the new velocity of the object?



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Sample Problem



Suppose a 1.5-kg brick is dropped on a glass table top from a height of 20 cm.

- a) What is the magnitude and direction of the impulse necessary to stop the brick?
- b) If the table top doesn't shatter, and stops the brick in 0.01 s, what is the average force it exerts on the brick?
- c) What is the average force that the brick exerts on the table top during this period?

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$$\begin{aligned} \text{a) } mgh &= \frac{1}{2}mv^2 \\ v &= \sqrt{2gh} \\ v &= \sqrt{(2)(9.8)(.2)} \\ v &= -1.98 \text{ m/s} \end{aligned}$$

$$\begin{aligned} p_i &= mv = 1.5\text{kg}(-1.98\text{m/s}) \\ &= -2.97 \text{ kg m/s} \\ \Delta p &= 0 - (-2.97 \text{ kg m/s}) \\ J &= \Delta p = 2.97 \text{ kg m/s} \end{aligned}$$

$$\begin{aligned} \text{b) } J &= Ft \\ F &= \frac{J}{t} = 297 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{c) } F &= -297 \text{ N} \\ & \text{(3rd Law)} \end{aligned}$$

Conservation of Momentum



Law of Conservation of Momentum

- If the resultant external force on a system is zero, then the vector sum of the momentums of the objects will remain constant.
- $\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$

Sample Problem



- A 75-kg man sits in the back of a 120-kg canoe that is at rest in a still pond. If the man begins to move forward in the canoe at 0.50 m/s relative to the shore, what happens to the canoe?

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$$\Sigma P_b = \Sigma P_a$$

$$0 = m_m v_m + m_c v_c$$

$$v_c = -\frac{m_m v_m}{m_c} = -\frac{(75 \text{ kg})(0.50 \text{ m/s})}{120 \text{ kg}}$$

$$v_c = -0.31 \text{ m/s}$$

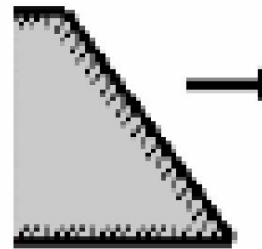
External versus internal forces

- External forces: forces coming from outside the system of particles whose momentum is being considered.
 - External forces change the momentum of the system.
- Internal forces: forces arising from interaction of particles within a system.
 - Internal forces cannot change momentum of the system.

An external force in **The System**

- The club head exerts an external impulsive force on the ball and changes its momentum.
- The acceleration of the ball is greater because its mass is smaller.

Clubhead



Golf Ball



The rightward-moving clubhead experiences a leftward force. The golf ball experiences a rightward force. The forces have equal magnitude and opposite direction.

An internal force in pool

The System

- The forces the balls exert on each other are internal and do not change the momentum of the system.
- Since the balls have equal masses, the magnitude of their accelerations is equal.



The rightward moving 7-ball experiences a leftward force. the 8-ball experiences a rightward force. The forces have equal magnitude and opposite directions.

Collision Types



- Elastic collisions
 - Also called “hard” collisions
 - No deformation occurs, no kinetic energy lost
- Inelastic collisions
 - Deformation occurs, kinetic energy is lost
- Perfectly Inelastic (stick together)
 - Objects stick together and become one object
 - Deformation occurs, kinetic energy is lost

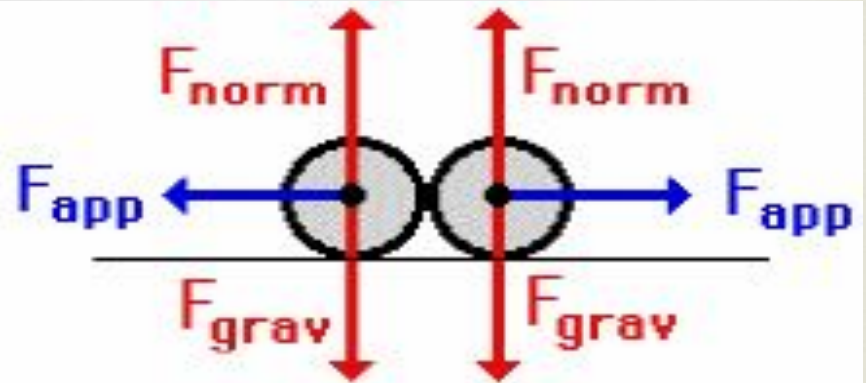
Collisions



- When two moving objects make contact with each other, they undergo a collision.
- Conservation of momentum is used to analyze all collisions.
- Newton's Third Law is also useful. It tells us that the force exerted by body A on body B in a collision is equal and opposite to the force exerted on body B by body A.

Collisions

- During a collision, external forces are ignored.
- The time frame of the collision is very short.
- The forces are impulsive forces (high force, short duration).



The collision between two billiard balls on a friction-free surface occurs in an isolated system; the only unbalanced forces occur from within the system.

Elastic Collision



- In elastic collisions, there is no deformation of colliding objects, and no change in kinetic energy of the system.
- Therefore, two basic equations must hold for all elastic collisions
 - $\Sigma p_b = \Sigma p_a$ (momentum conservation)
 - $\Sigma K_b = \Sigma K_a$ (kinetic energy conservation)

(Perfectly) Inelastic Collisions

- Simplest type of collisions.
- After the collision, there is only one velocity, since there is only one object.
- Kinetic energy is lost.
- Explosions are the reverse of perfectly inelastic collisions in which kinetic energy is gained!

Explosions



- When an object separates suddenly, as in an explosion, all forces are internal.
- Momentum is therefore conserved in an explosion.
- There is also an increase in kinetic energy in an explosion. This comes from a potential energy decrease due to chemical combustion.

Recoil



- Guns and cannons “recoil” when fired.
- This means the gun or cannon must move backward as it propels the projectile forward.
- The recoil is the result of action-reaction force pairs, and is entirely due to **internal forces**. As the gases from the gunpowder explosion expand, they push the projectile forwards and the gun or cannon backwards.

Sample problem



-
- Suppose a 5.0-kg projectile launcher shoots a 209 gram projectile at 350 m/s. What is the recoil velocity of the projectile launcher?

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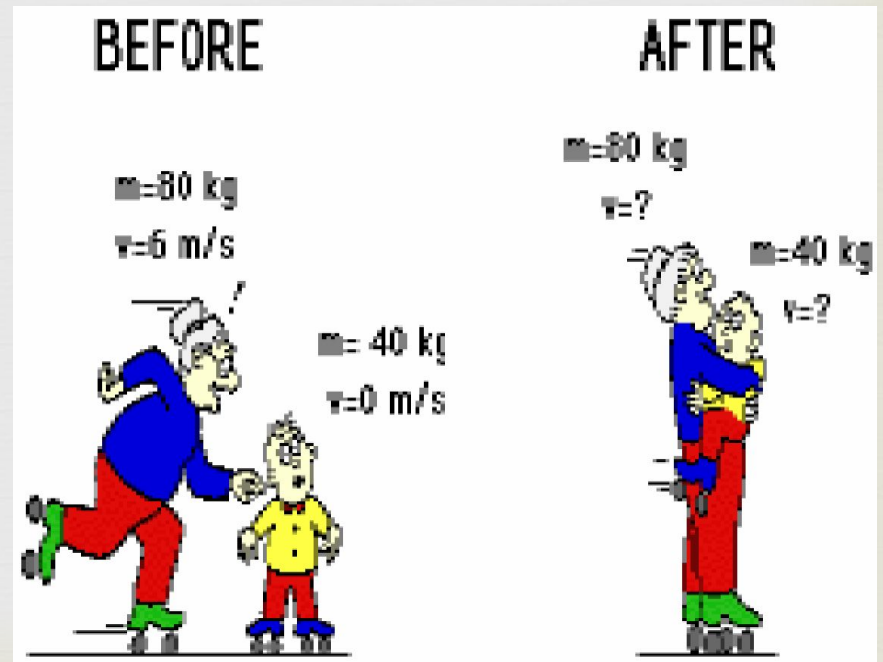
$$\sum P_B = \sum P_A$$

$$0 = m_L V_L + m_P V_P$$

$$V_L = - \frac{m_P V_P}{m_L} = - \frac{(0.209 \text{ kg})(350 \text{ m/s})}{5 \text{ kg}}$$
$$= - 14.63 \text{ m/s}$$

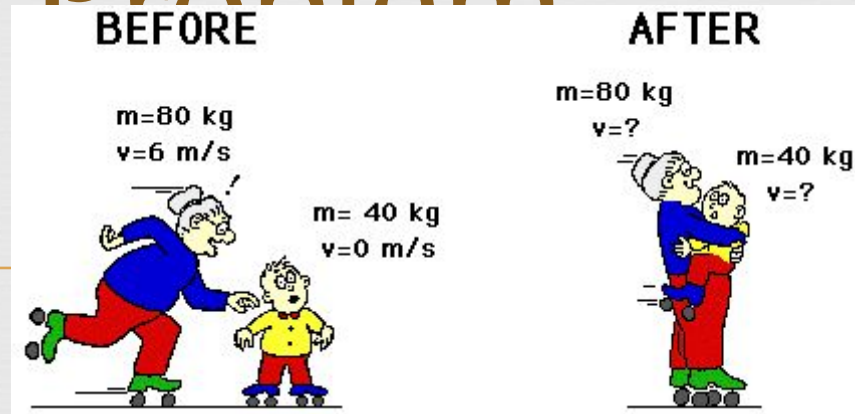
Sample Problem

- An 80-kg roller skating grandma collides inelastically with a 40-kg kid. What is their velocity after the collision?
- How much kinetic energy is lost?



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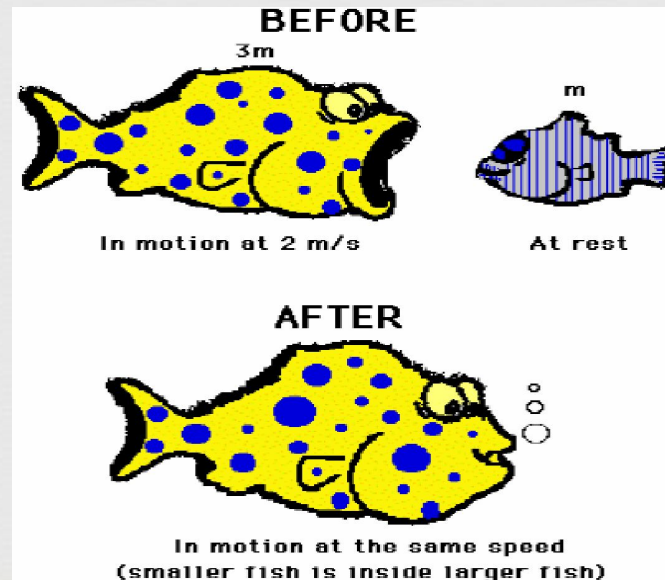
$$\begin{aligned}
 P_B &= P_A \\
 m_G v_{GB} + 0 &= (m_G + m_K) v_{GA} \\
 (80 \text{ kg})(6 \text{ m/s}) &= (120 \text{ kg}) v_{GA} \\
 v_{GA} &= 4.0 \text{ m/s}
 \end{aligned}$$

Before	After
$\frac{1}{2} m_G v_G^2$	$\frac{1}{2} (m_G + m_K) (v_{GA})^2$
$\frac{1}{2} (80 \text{ kg}) (6 \text{ m/s})^2$	$\frac{1}{2} (120 \text{ kg}) (4 \text{ m/s})^2$
1440 J	960 J

$$\Delta K = 960 \text{ J} - 1440 \text{ J} = \boxed{-480 \text{ J}}$$

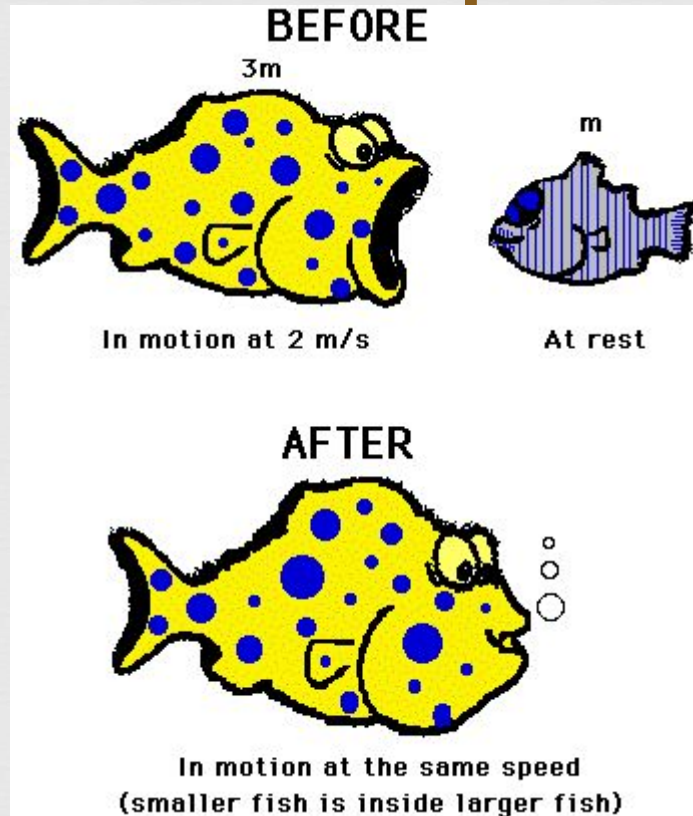
Sample Problem

- A fish moving at 2 m/s swallows a stationary fish which is $1/3$ its mass. What is the velocity of the big fish after dinner?



Sample Problem

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Before		after
$(3m) v_B$	$=$	$(4m) v_A$
$(3m)(2\text{ m/s})$	$=$	$(4m) v_A$

$$v_A = \frac{6\text{ m/s}}{4} = 1.5\text{ m/s}$$

Sample Problem



- A 500-g cart moving at 2.0 m/s on an air track elastically strikes a 1,000-g cart at rest. What are the resulting velocities of the two carts?

2D-Collisions



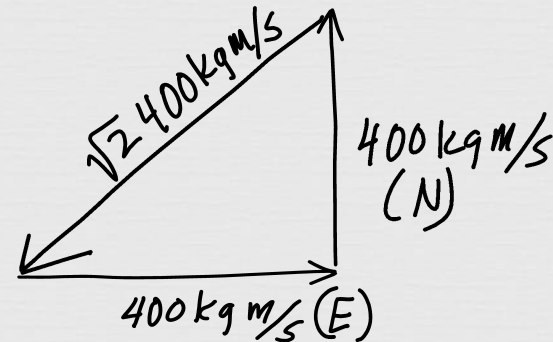
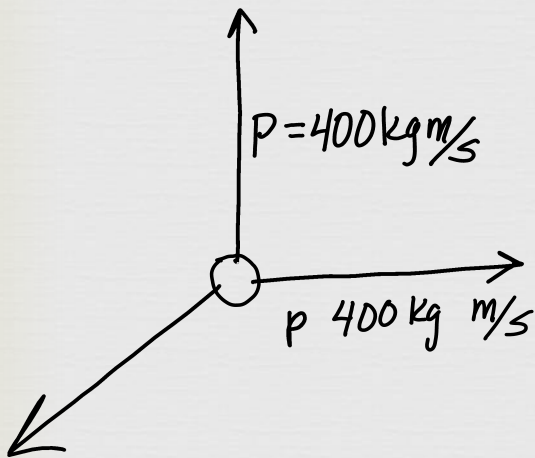
- Momentum in the x-direction is conserved.
 - ΣP_x (before) = ΣP_x (after)
- Momentum in the y-direction is conserved.
 - ΣP_y (before) = ΣP_y (after)
- Treat x and y coordinates independently.
 - Ignore x when calculating y
 - Ignore y when calculating x
- Let's look at a simulation:
 - <http://surendranath.tripod.com/Applets.html>

Sample Problem

- An exploding object breaks □ into three fragments. A 2.0 kg fragment travels north at 200 m/s. A 4.0 kg fragment travels east at 100 m/s. The third fragment has mass 3.0 kg. What is the magnitude and direction of its velocity?

Sample Problem

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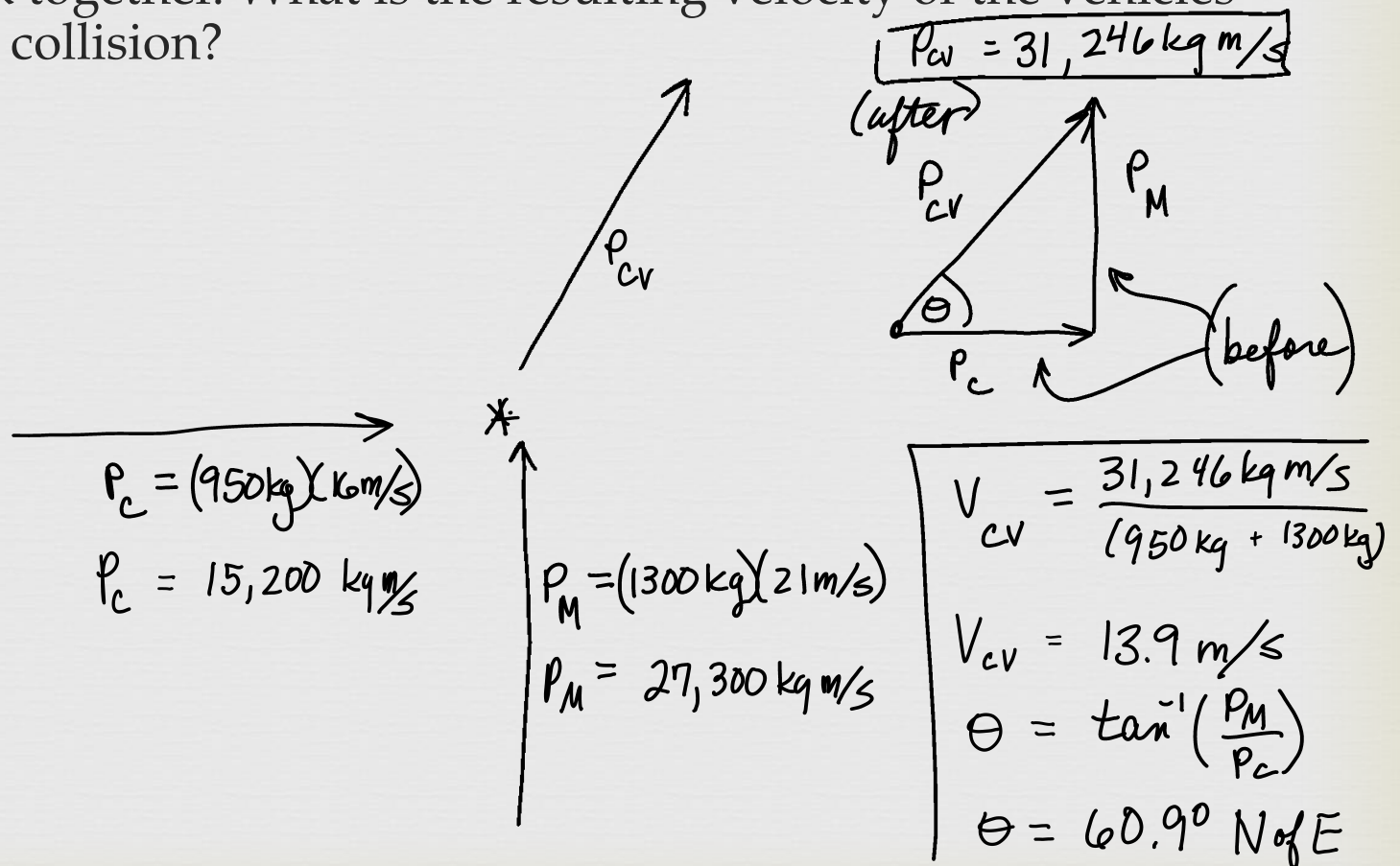


Sample problem

- A car with a mass of 950 kg and a speed of 16 m/s to the east approaches an intersection. A 1300-kg minivan traveling north at 21 m/s approaches the same intersection. The vehicles collide and stick together. What is the resulting velocity of the vehicles after the collision?

Sample problem

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Sample Problem

- A 500-g cart moving at 2.0 m/s on an air track elastically strikes a 1,000-g cart at rest. What are the resulting velocities of the two carts?

$$\begin{aligned} \sum P_B &= \sum P_A \\ m_1 v_{1B} + m_2 v_{2B} &= m_1 v_{1A} + m_2 v_{2A} \\ (0.5)(2) &= (0.5)(v_{1A}) + v_{2A} \\ \boxed{1 = 0.5x + y} \end{aligned}$$

$$\begin{aligned} \sum K_B &= \sum K_A \\ \cancel{\frac{1}{2} m_1 v_{1B}^2} + \cancel{\frac{1}{2} m_2 v_{2B}^2} &= \cancel{\frac{1}{2} m_1 v_{1A}^2} + \cancel{\frac{1}{2} m_2 v_{2A}^2} \\ 0.5(2)^2 &= 0.5v_{1A}^2 + v_{2A}^2 \\ \boxed{2 = 0.5x^2 + y^2} \end{aligned}$$

SOLVE $(1 = 0.5x + y \text{ AND}$

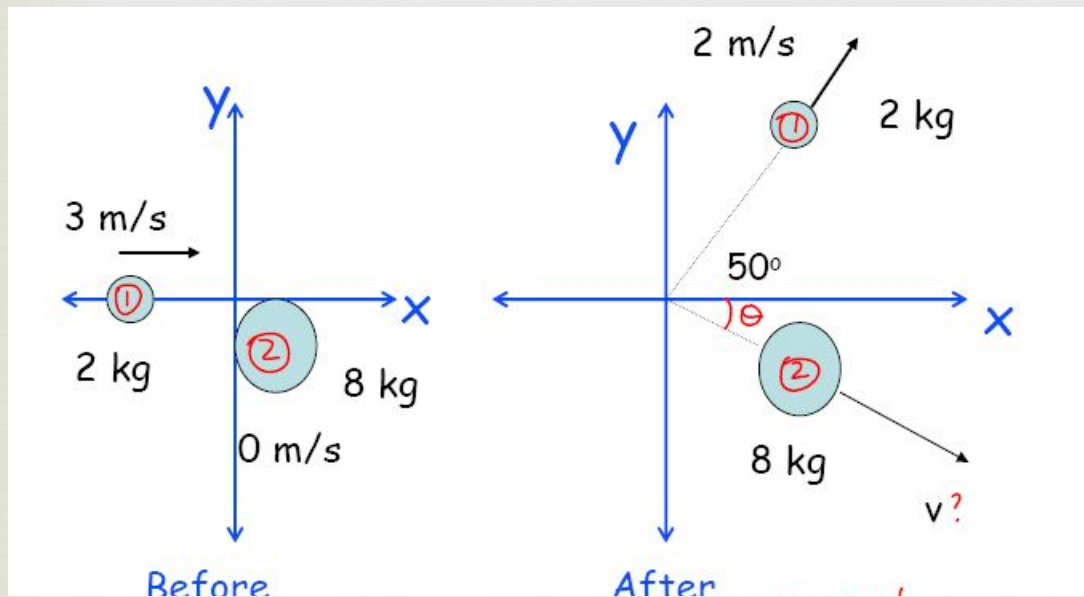
$2 = 0.5x^2 + y^2, x)$

soln 1 $(x, y) = (-\frac{2}{3}, \frac{4}{3}) \text{ m/s}$

soln 2 $(x, y) = (2, 0) \text{ m/s}$

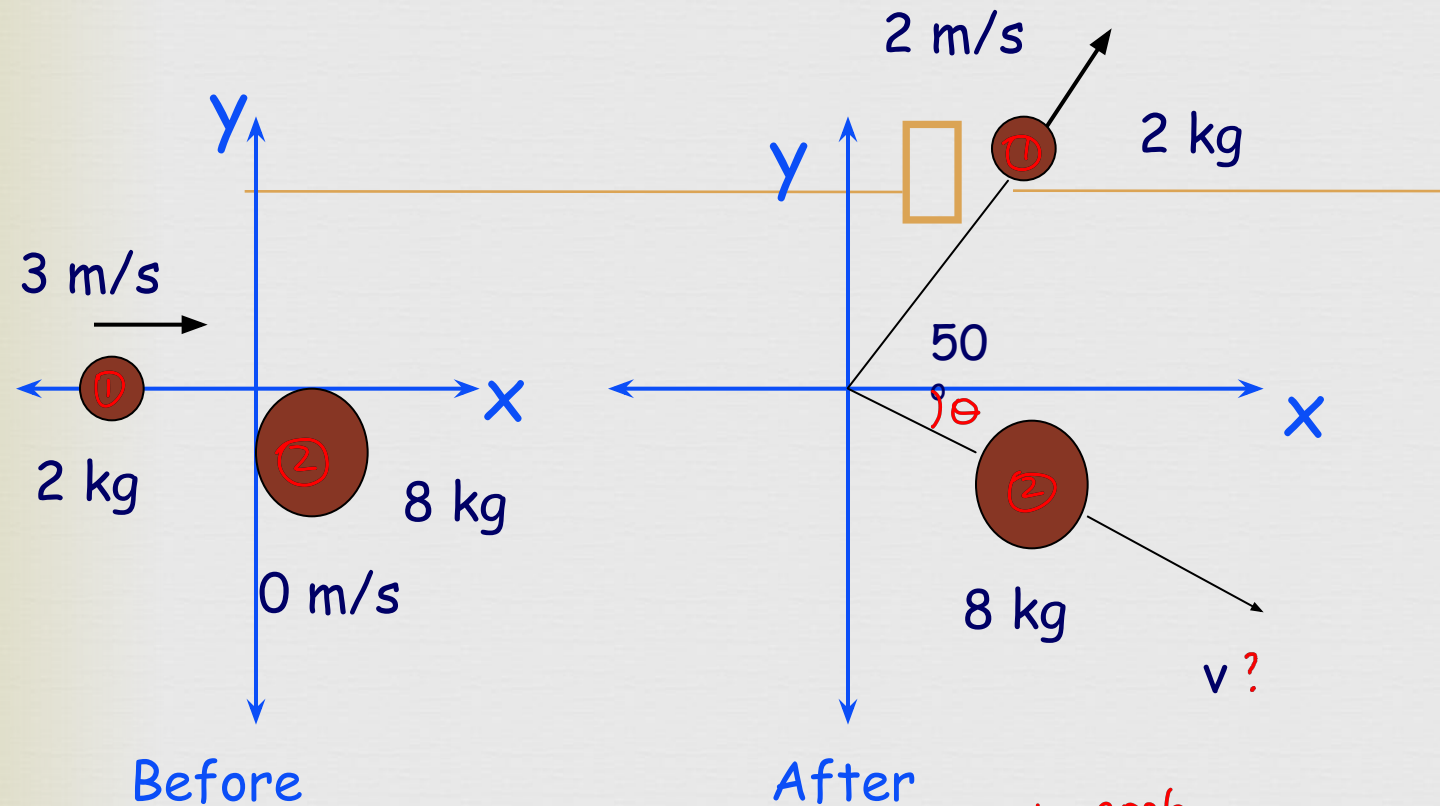
Sample problem

□ Calculate velocity of 8-kg ball after the collision.



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$$V = \sqrt{(.43)^2 + (-.39)^2}$$

$$V = 0.58 \text{ m/s}$$

$$\Theta = \tan^{-1}\left(\frac{-.39}{.43}\right)$$

$$\theta = -42.2^\circ$$

Before
x-prob

$$\sum P_B = \sum P_A$$

$$m_1 V_{1B} = m_1 V_{1AX} + m_2 V_{2AX}$$

$$(2)(3) = (2)(2 \cos 50^\circ) + 8 V_{2AX}$$

$$V_{2AX} = 0.43 \text{ m/s}$$

y-prob

$$\sum P_B = \sum P_y$$

$$0 = m_1 V_{1AY} + m_2 V_{2AY}$$

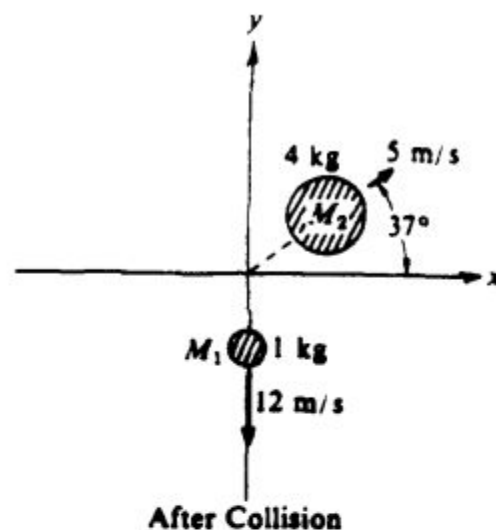
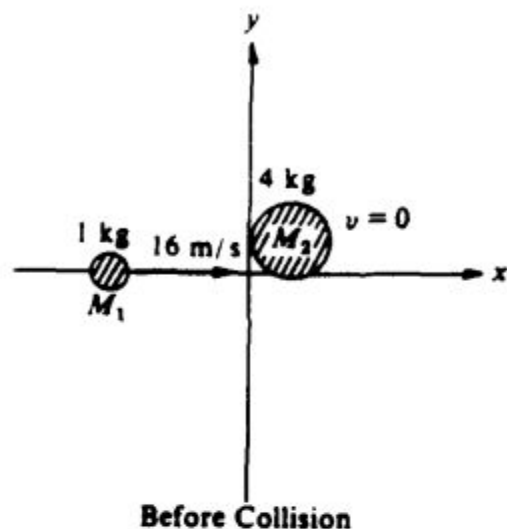
$$0 = (2)(2 \sin 50^\circ) + 8 V_{2AY}$$

$$V_{2AY} = -0.39 \text{ m/s}$$

Sample Problem



-
- Suppose three equally strong, equally massive astronauts decide to play a game as follows: The first astronaut throws the second astronaut towards the third astronaut and the game begins. Describe the motion of the astronauts as the game proceeds. Assume each toss results from the same-sized "push." How long will the game last?



View From Above

1984B2. Two objects of masses $M_1 = 1$ kilogram and $M_2 = 4$ kilograms are free to slide on a horizontal frictionless surface. The objects collide and the magnitudes and directions of the velocities of the two objects before and after the collision are shown on the diagram above. ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $\tan 37^\circ = 0.75$)

- a. Calculate the x and y components (p_x and p_y , respectively) of the momenta of the two objects before and after the collision, and write your results in the proper places in the following table.

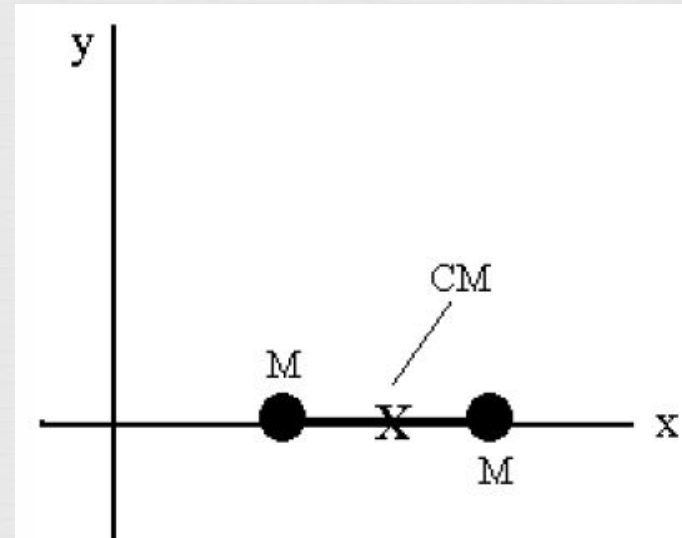
	$M_1 = 1 \text{ kg}$		$M_2 = 4 \text{ kg}$	
	$p_x \left(\frac{\text{kg} \cdot \text{m}}{\text{s}} \right)$	$p_y \left(\frac{\text{kg} \cdot \text{m}}{\text{s}} \right)$	$p_x \left(\frac{\text{kg} \cdot \text{m}}{\text{s}} \right)$	$p_y \left(\frac{\text{kg} \cdot \text{m}}{\text{s}} \right)$
Before Collision				
After Collision				

- b. Show, using the data that you listed in the table, that linear momentum is conserved in this collision.
 c. Calculate the kinetic energy of the two-object system before and after the collision.
 d. Is kinetic energy conserved in the collision?

CENTER OF MASS

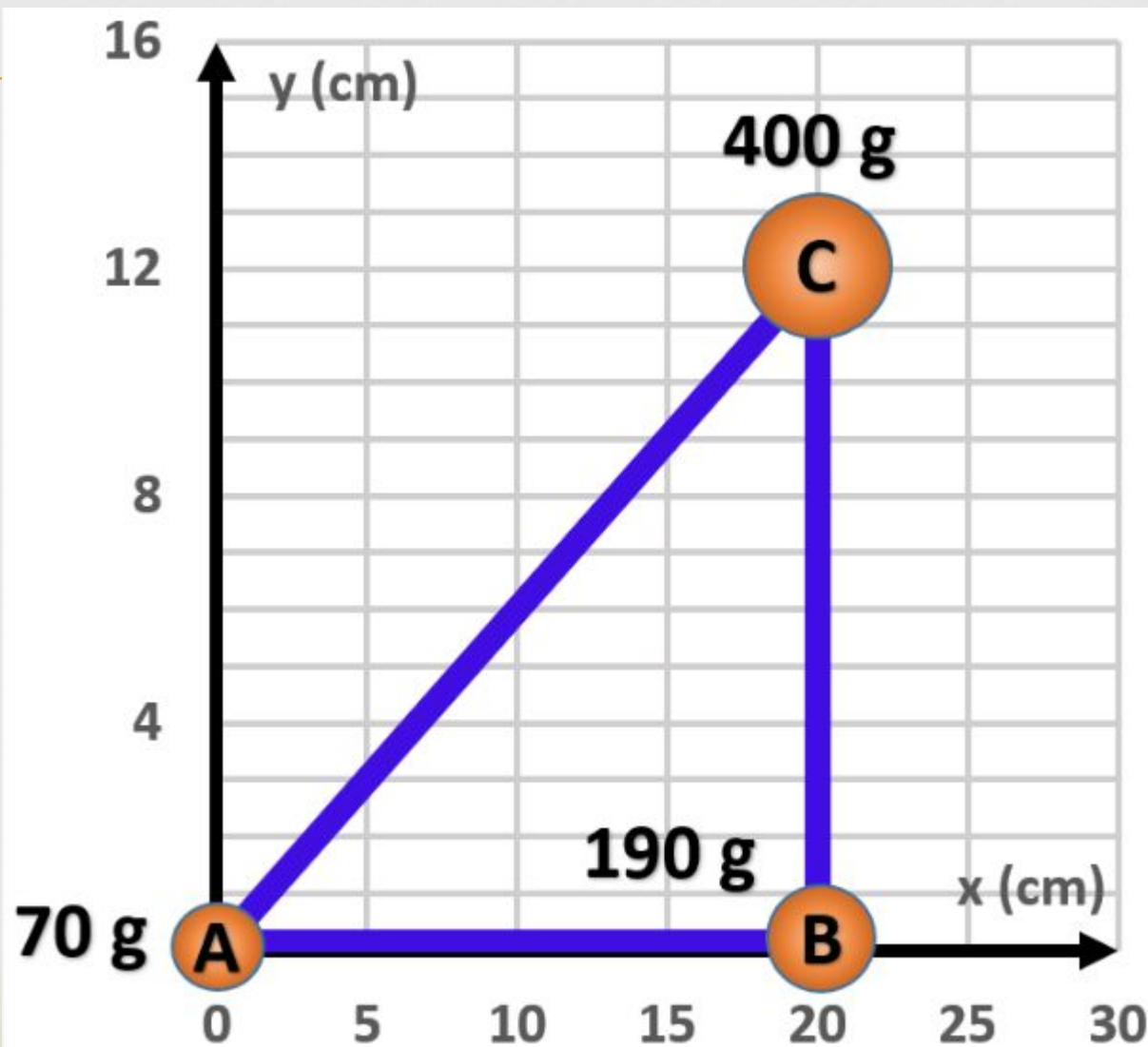
Up to the point in the course we have treated everything as a single point in space no matter how much mass it has. We call this "Special point" the **CENTER OF MASS**. The **CM** is basically a coordinate which represents the average position of mass.

Consider an extended rigid body made up of 2 or more point masses, such as a dumb-bell. We want to treat this as a system, but we don't want to just assume that the CM is in the middle (even though we know it is).



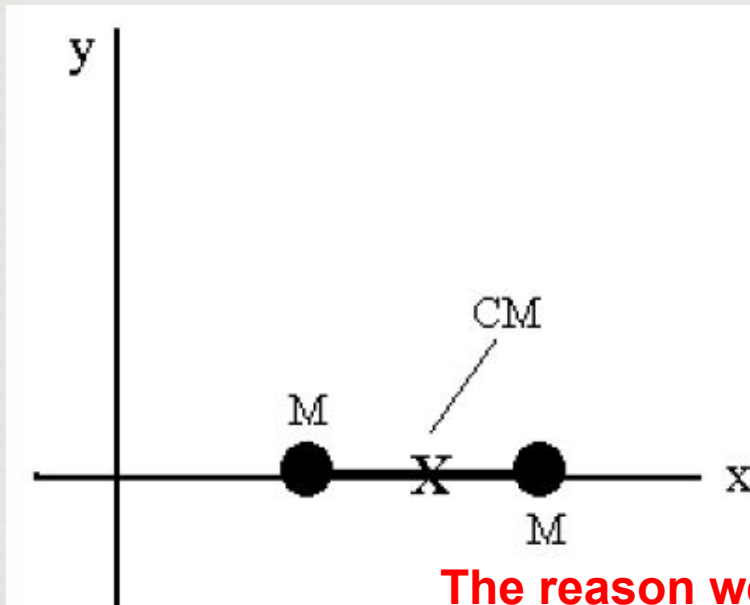
The reason we are concerned here is because if this system were to move, how would you determine the momentum? Would you just assume ALL the mass was in the middle and treat it as a point?

Calculate the center of mass for this arrangement of masses



Momentum of point masses

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